[COLLEGE NAME]

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DEPARTMENT OF ELECTRONICS

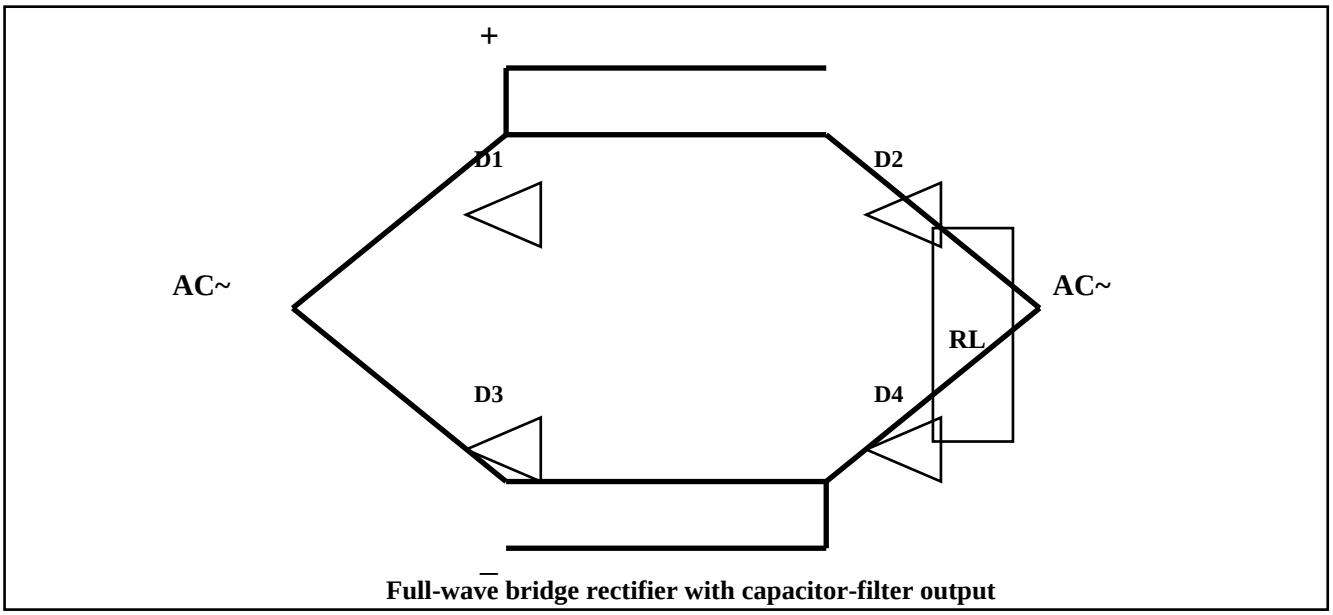
PROJECT REPORT

ON

"FULL WAVE RECTIFIER WITH CAPACITOR FILTER"

Project Guide:
[PROJECT GUIDE NAME]

Submitted By:
1. [STUDENT NAME]
2. [STUDENT NAME]
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Full-wave bridge rectifier with capacitor-filter output

AC to DC conversion using a four-diode bridge and capacitor filter

CERTIFICATE

This is to certify that the project entitled **"FULL WAVE RECTIFIER WITH CAPACITOR FILTER"** has been completed by the students of the Department of Electronics under the guidance of the project guide during the academic year **2026–2027**.

Student Names:
1. _____
2. _____
3. _____

Project Guide Signature: _____

Principal Signature and Seal: _____

Internal Examiner: _____

External Examiner: _____

Place: _____ Date: _____

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide for valuable guidance, technical suggestions, and continuous encouragement. We thank the Principal, the Department of Electronics, and the laboratory staff for providing the facilities required to complete this project.

We are also thankful to our teachers, classmates, friends, and family members for their support. This project helped us understand the transformer, diode bridge, capacitor filter, load resistor, LED indicator, and measurement instruments.

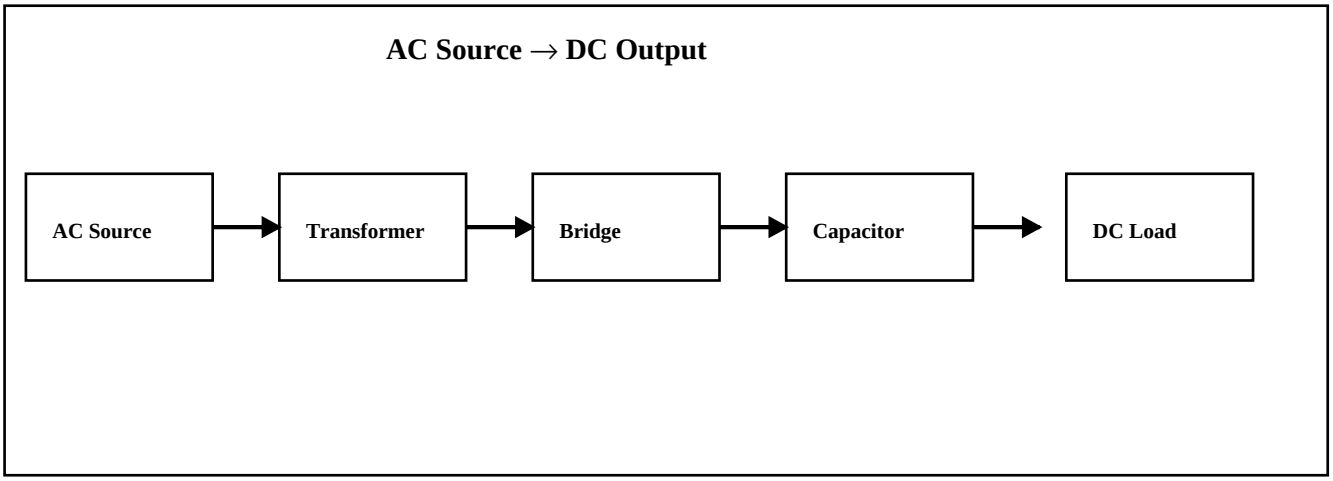
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PLACE: _____

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PROJECT IDEA / OVERVIEW



The project converts low-voltage alternating current into direct current. The transformer reduces the input voltage, the four diodes perform full-wave rectification, and the capacitor filter smooths the pulsating output.

The load resistor represents the circuit being powered. The LED provides a visual indication that DC voltage is present. The complete model demonstrates the basic stages of a simple power supply.

ABSTRACT

A rectifier is an electronic circuit used to convert AC into DC. Many electronic circuits require a fixed polarity supply, so rectification is an important stage in adapters, chargers, and power supplies.

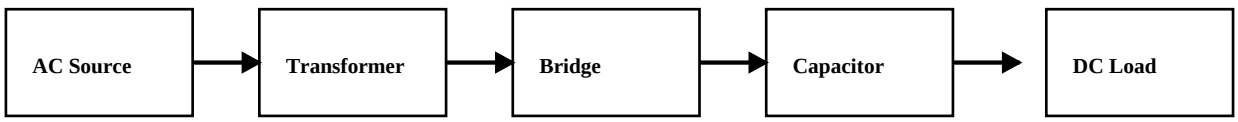
This project uses four PN junction diodes in a bridge arrangement. Two diodes conduct during each half cycle, so the load current flows in the same direction for both halves of the AC waveform.

A capacitor connected across the load charges near the waveform peaks and discharges between peaks. This reduces ripple and provides smoother DC. The project also develops practical skills in wiring, measurement, polarity checking, and electrical safety.

INTRODUCTION

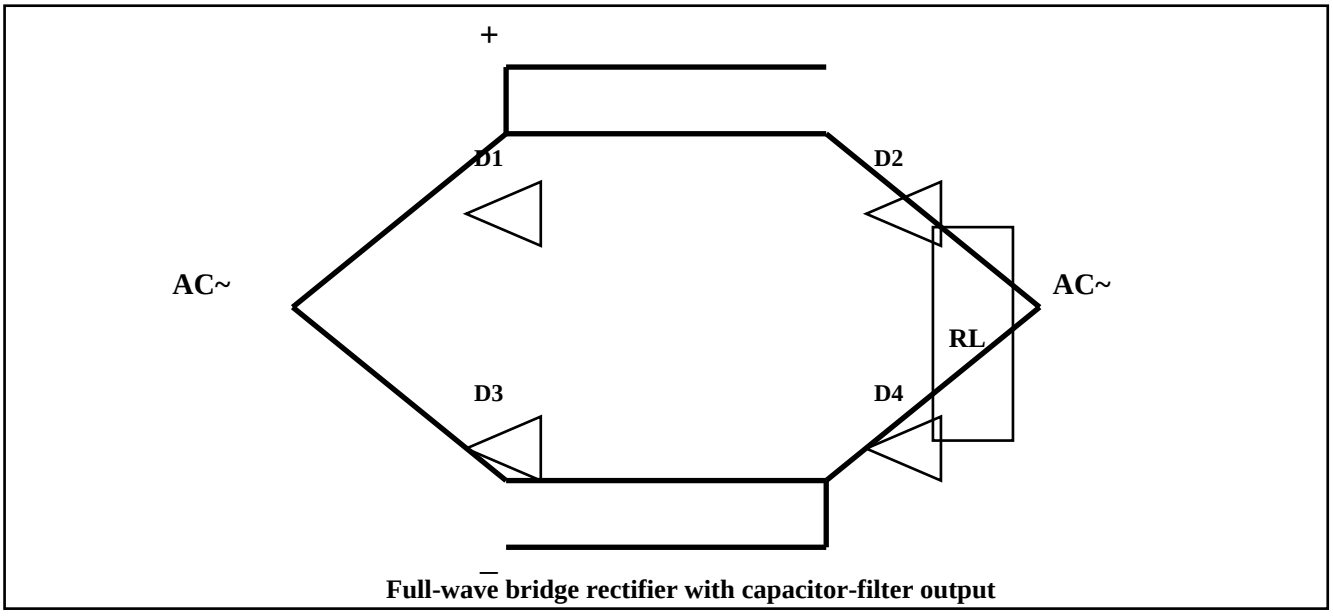
AC: Alternating current changes magnitude and direction periodically.
DC: Direct current flows in one direction and has fixed polarity.
Rectifier: A diode circuit that converts AC into unidirectional current.
Full-wave bridge rectifier: A four-diode circuit that uses both AC half cycles.
Capacitor filter: A parallel capacitor that reduces output ripple.

AC → Transformer → Bridge → Filter → Load



Applications include low-voltage power supplies, battery chargers, LED circuits, adapters, audio circuits, and laboratory demonstrations.

BASIC CIRCUIT DIAGRAM



Full-wave bridge rectifier with capacitor-filter output

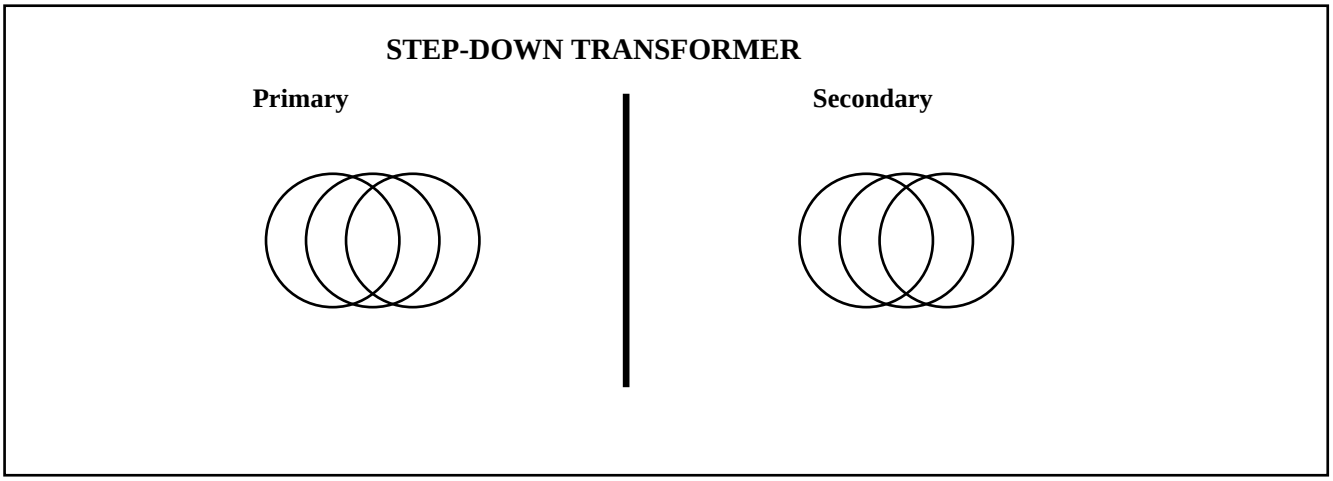
The transformer secondary is connected to the two AC terminals of the bridge. The positive and negative bridge terminals provide pulsating DC. The capacitor and load are connected in parallel across the DC output.

The LED is connected through a current-limiting resistor. Correct diode orientation, capacitor polarity, and LED polarity must be checked before power is applied.

COMPONENTS REQUIRED

SR. NO.	COMPONENT	QTY.	PURPOSE
1	Step-down transformer	1	Reduces AC voltage
2	PN junction diodes	4	Bridge rectification
3	Electrolytic capacitor	1	Ripple filtering
4	Load resistor	1	Represents load
5	LED + series resistor	1 set	Output indication
6	Breadboard / PCB	1	Mounting
7	Connecting wires	As required	Connections
8	Low-voltage AC source	1	Supply
9	Multimeter	1	Measurement
10	Switch / accessories	Optional	Control and mounting

STEP-DOWN TRANSFORMER

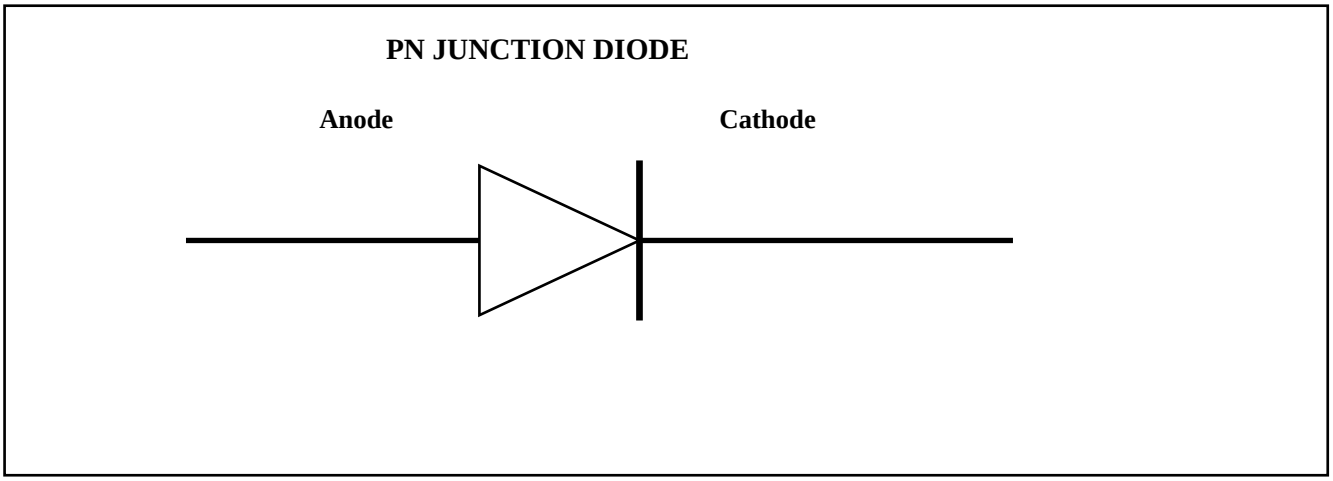


A transformer transfers AC energy through mutual induction. A step-down transformer has more turns on the primary than on the secondary, producing a lower secondary voltage.

Function in this project: It supplies safe low-voltage AC to the bridge. It does not convert AC into DC.

Important information: Typical educational secondary ratings are 6–12 V AC, depending on the required output.

PN JUNCTION DIODE

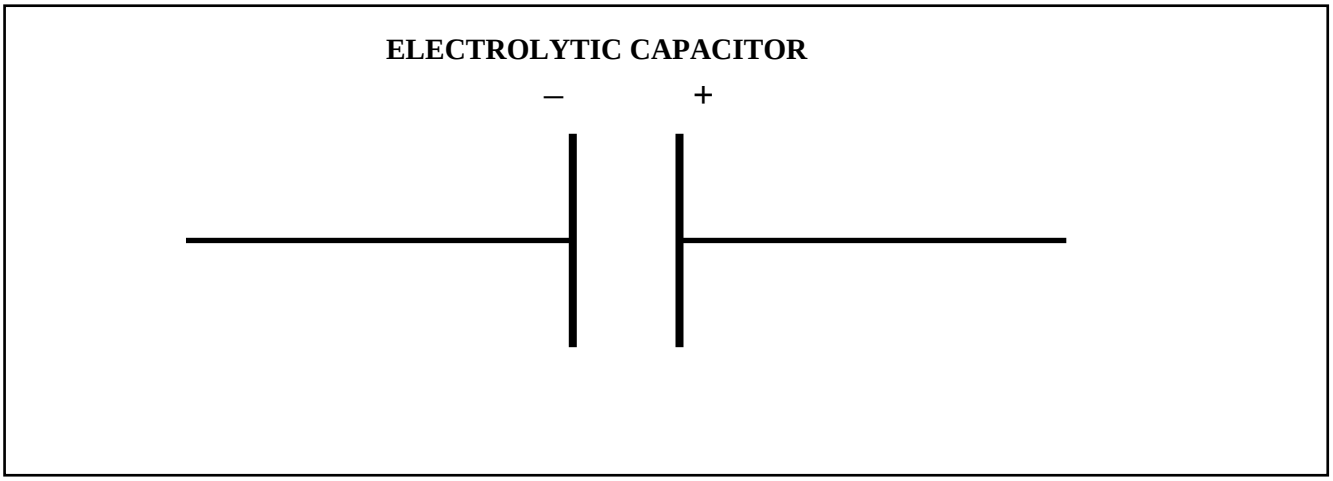


A diode conducts mainly in one direction. It conducts in forward bias and blocks current in reverse bias within its rating.

Function in this project: Four diodes form the bridge. Two conduct in each half cycle and keep load polarity unchanged.

Important information: 1N4007 is a common low-frequency example. Actual forward drop and reverse rating must be checked.

ELECTROLYTIC CAPACITOR

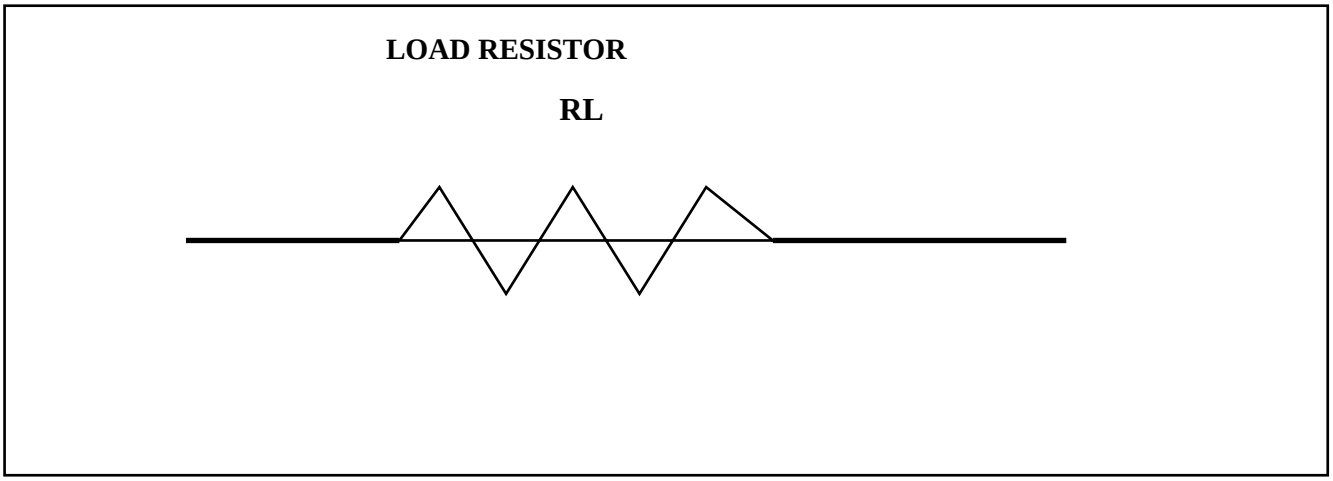


A capacitor stores charge. An electrolytic capacitor is polarized and must be connected correctly.

Function in this project: It charges near peaks and discharges through the load, reducing ripple.

Important information: Capacitance is measured in µF and voltage rating must exceed the expected DC voltage. Q = CV.

LOAD RESISTOR

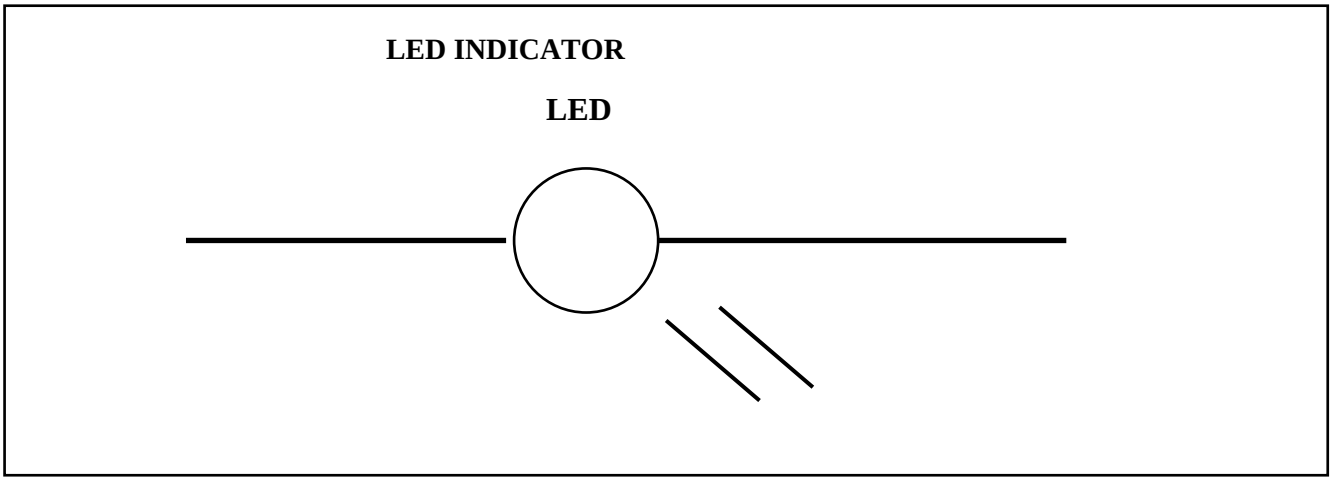


A resistor opposes current and dissipates electrical energy as heat.

Function in this project: It represents the load and provides a discharge path for the capacitor.

Important information: Ohm’s law: V = IR. Power: P = VI. Select a suitable wattage rating.

LED INDICATOR

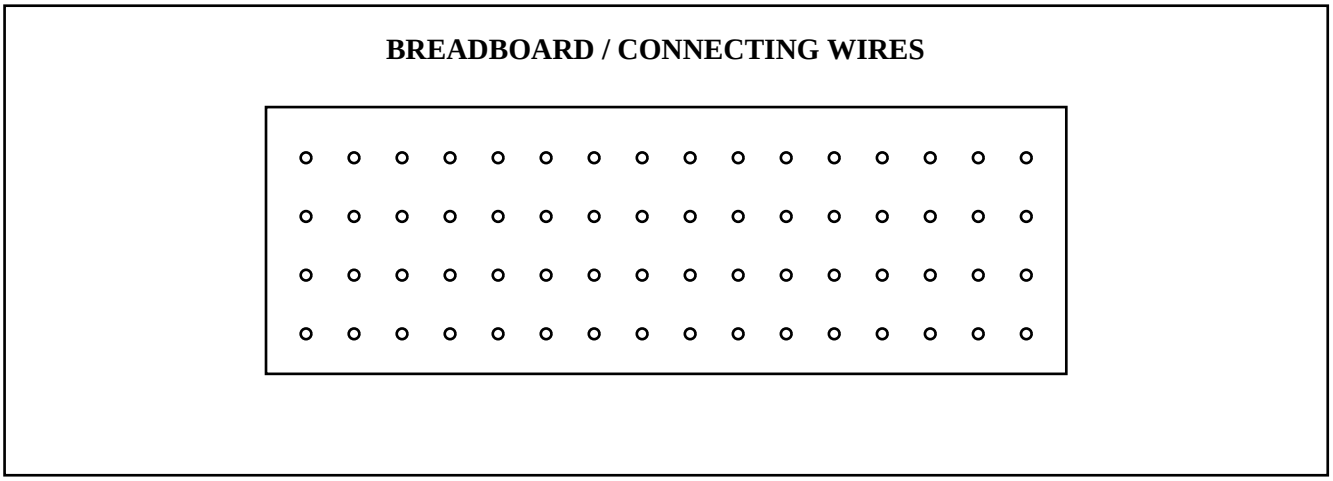


An LED emits light when forward current flows through its semiconductor junction.

Function in this project: It indicates the presence of DC output and must be connected with a series current-limiting resistor.

Important information: Observe anode/cathode polarity. Never connect an LED directly across the supply.

BREADBOARD / CONNECTING WIRES



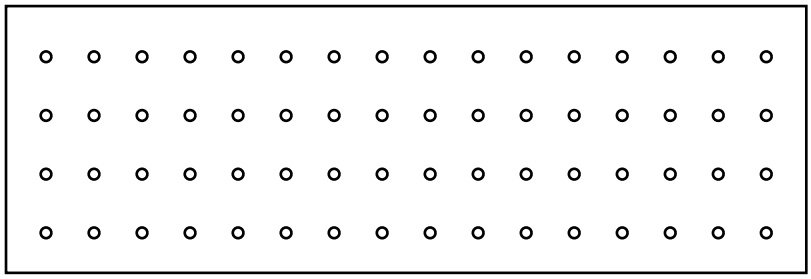
A breadboard permits temporary construction without soldering. Its holes are internally connected in groups.

Function in this project: Wires connect the transformer, bridge, capacitor, resistor, and LED.

Important information: Check rows, continuity, loose wires, and accidental short circuits.

AC SOURCE / MULTIMETER

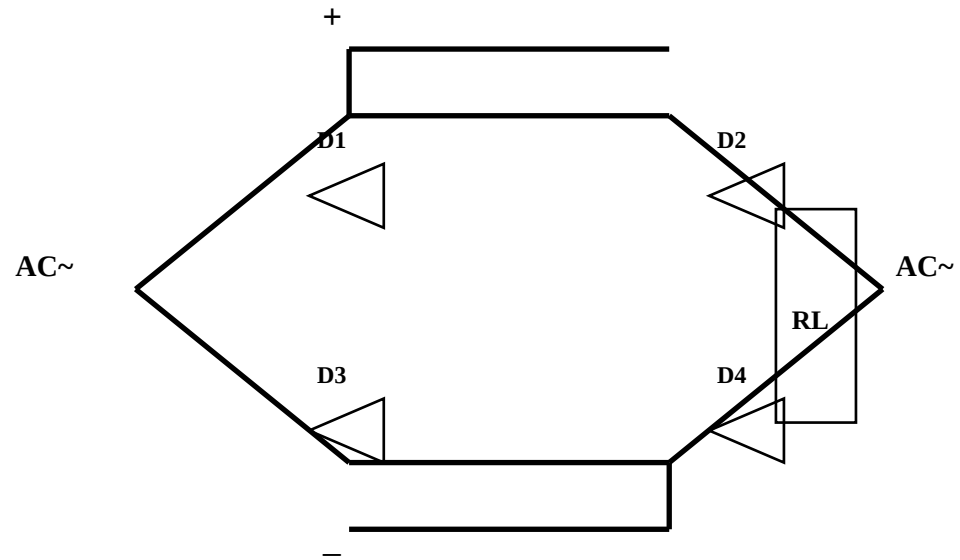
AC Source and Multimeter



A low-voltage isolated AC source supplies the transformer or its secondary winding. A digital multimeter measures AC voltage, DC voltage, resistance, and continuity. An oscilloscope may be used to observe waveforms.

Never connect the breadboard directly to 230 V AC mains. Switch off power before changing wiring and use the correct meter range.

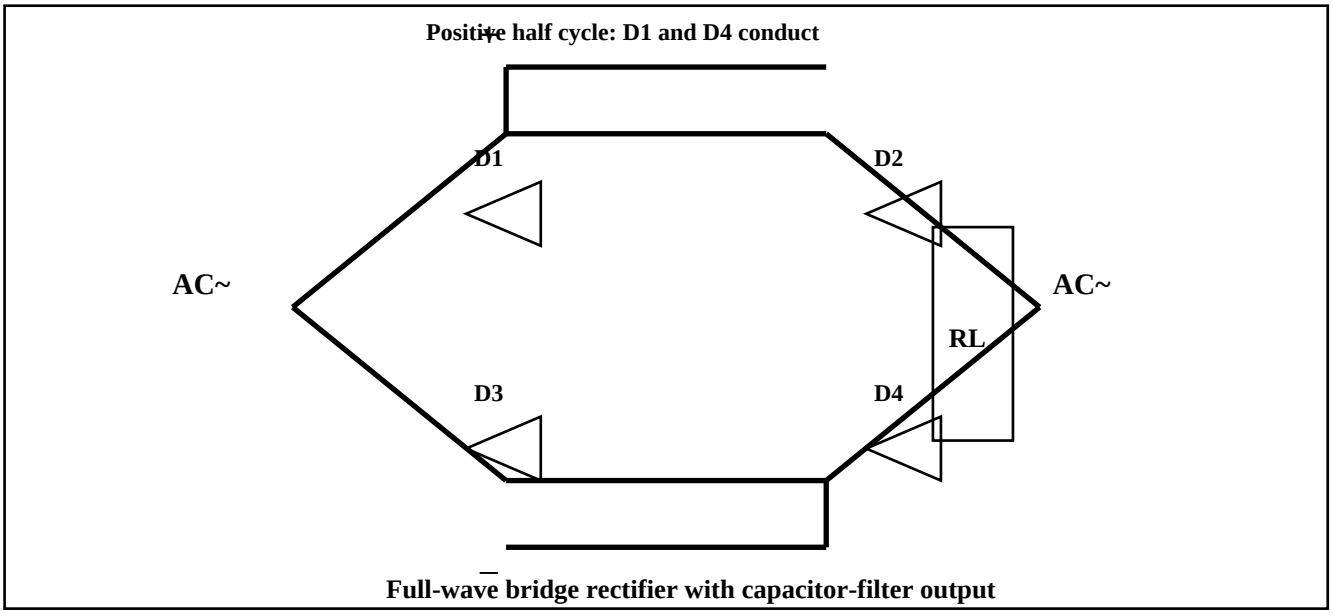
WORKING PRINCIPLE



Full-wave bridge rectifier with capacitor-filter output

1. AC supply is applied to the transformer.
2. The transformer reduces the voltage.
3. Four diodes form the bridge.
4. Two diodes conduct during one half cycle.
5. The alternate pair conducts during the other half cycle.
6. Load current remains in the same direction.
7. The output is pulsating DC.
8. The capacitor reduces ripple.
9. The LED indicates output voltage.

POSITIVE HALF CYCLE

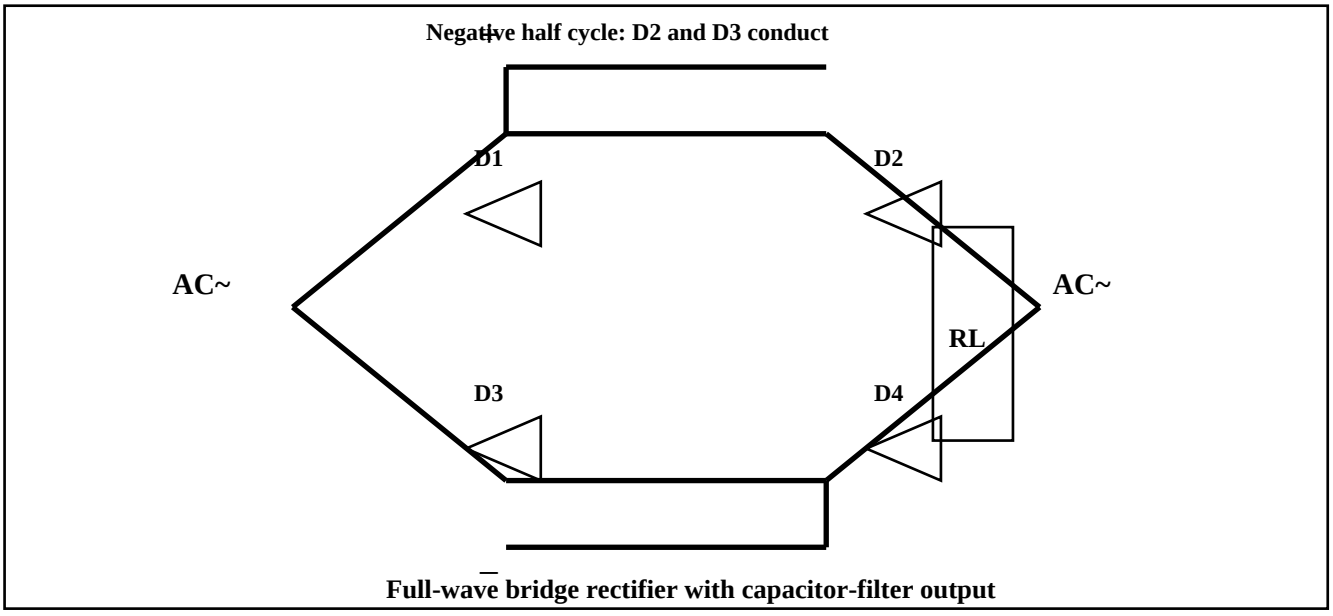


Full-wave bridge rectifier with capacitor-filter output

During the positive half cycle, the conducting pair is D1 and D4 in the labelled arrangement. D2 and D3 are reverse biased. Current flows through the load in the forward direction and the capacitor charges toward the peak voltage.

Current path: AC terminal → D1 → positive output → load/capacitor branch → negative output → D4 → other AC terminal.

NEGATIVE HALF CYCLE

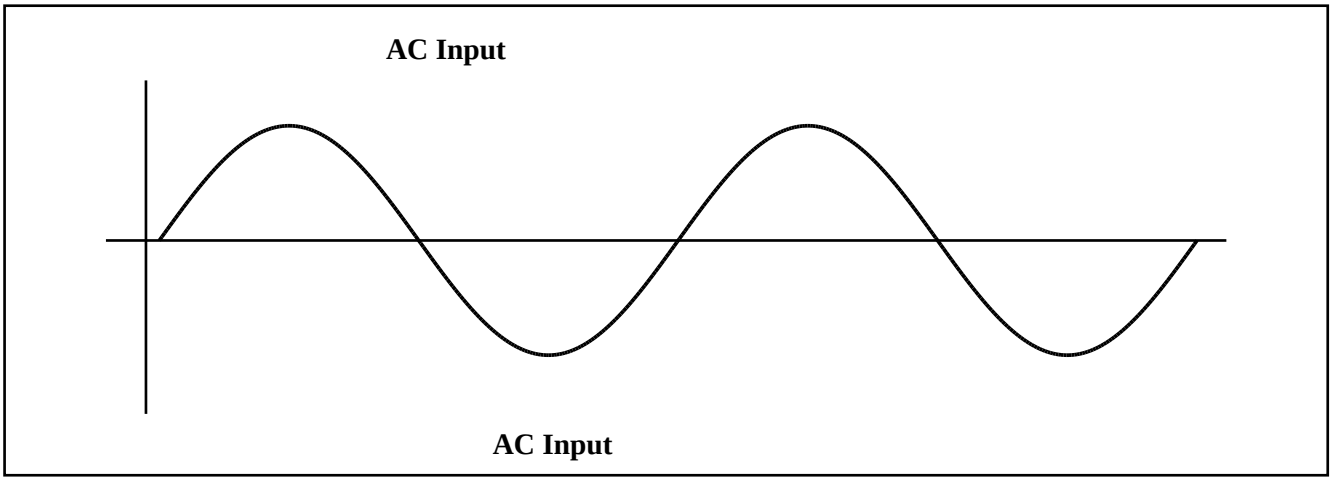


Full-wave bridge rectifier with capacitor-filter output

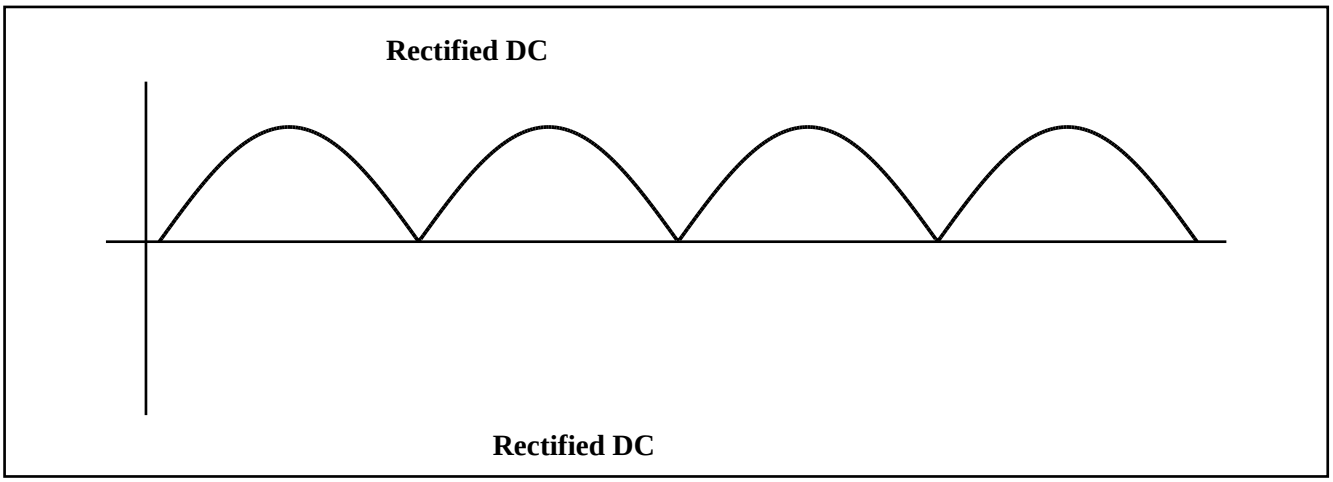
During the negative half cycle, D2 and D3 conduct while D1 and D4 are reverse biased. The bridge changes the current path but maintains the same polarity across the load.

Current path: Alternate AC terminal → D2 → positive output → load/capacitor branch → negative output → D3 → other AC terminal.

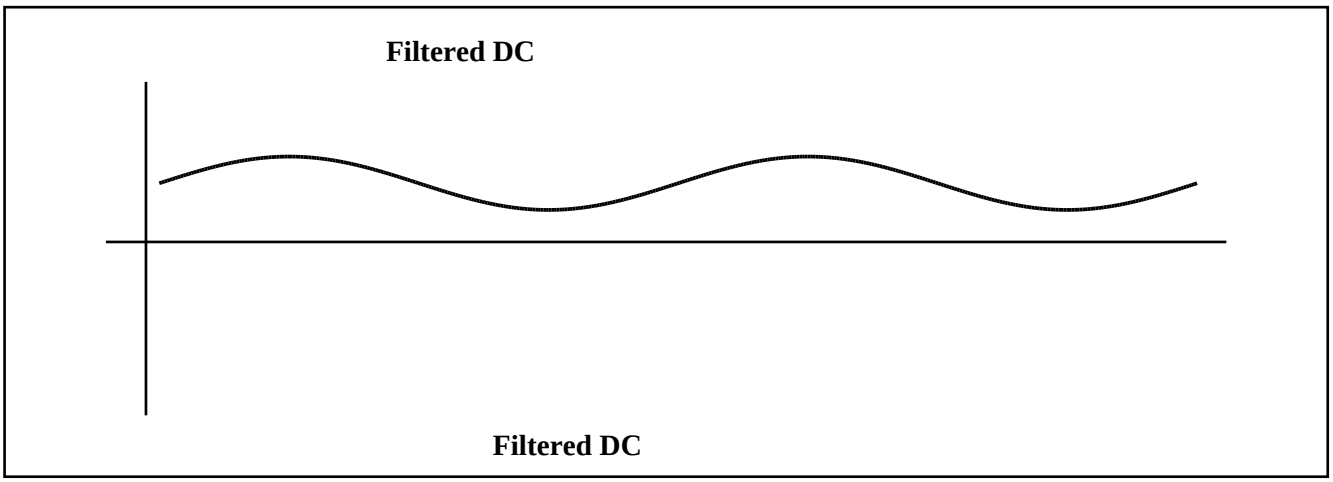
OUTPUT WAVEFORM



The input alternates above and below zero.

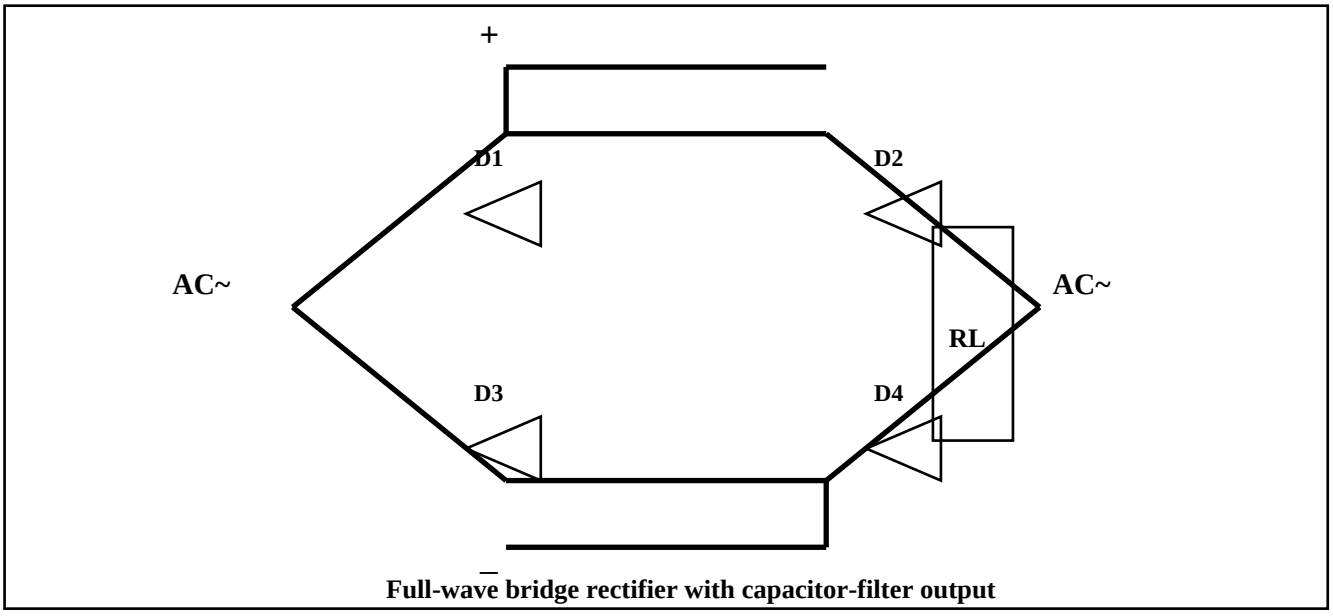


Both half cycles appear with the same output polarity.



The capacitor reduces the variation between peaks.

CAPACITOR FILTER



Full-wave bridge rectifier with capacitor-filter output

The capacitor is connected across the DC output. It charges when the rectified voltage rises above its stored voltage. When the rectified voltage falls, the capacitor discharges through the load and supplies current between peaks.

Increasing capacitance generally reduces ripple, but it can increase the initial charging current. The capacitor must have correct polarity and a suitable voltage rating.

Ripple frequency: $f_{ripple} = 2f_{AC}$
Approximate ripple: $V_{r(pp)} \approx I_L / (f_{ripple} C)$

RIPPLE AND EFFICIENCY

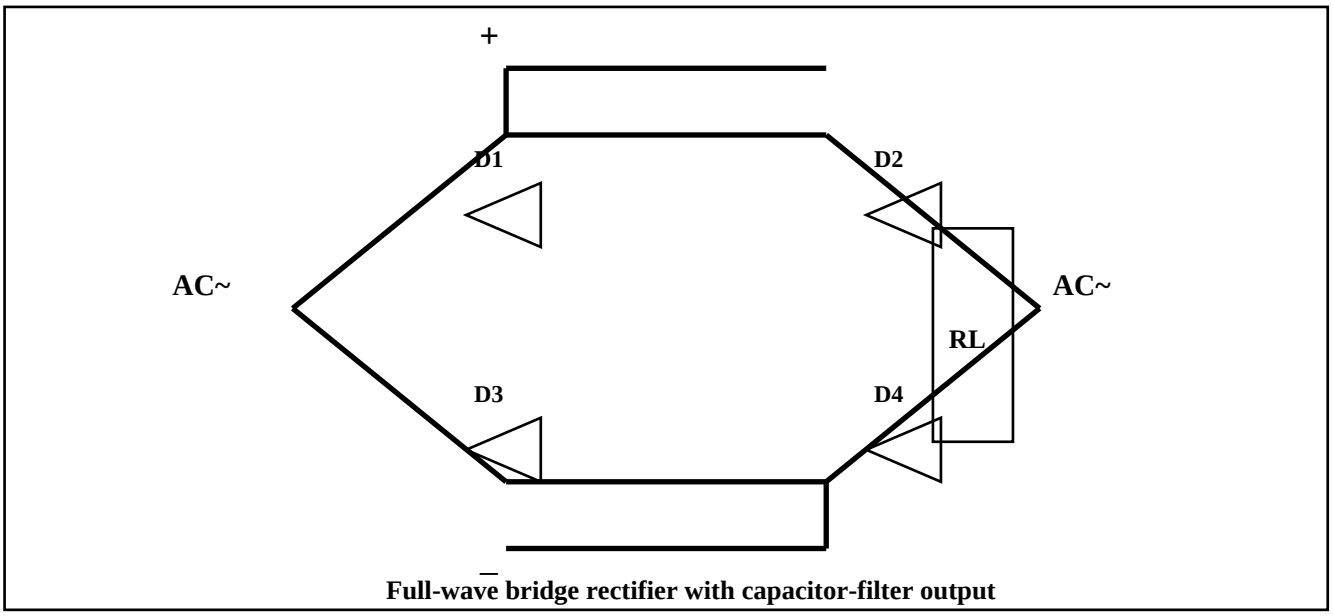
Ripple is the unwanted AC variation remaining in the DC output. It depends on load current, capacitance, input frequency, transformer voltage, diode drops, and capacitor losses.

The theoretical maximum rectification efficiency of a full-wave rectifier is approximately **81.2%** without additional practical losses. Actual efficiency is lower because of diode forward voltage drops, transformer losses, wiring resistance, heating, and capacitor ESR.

A capacitor filter reduces ripple but does not eliminate it completely. A voltage regulator may be added when a stable DC voltage is required.

CONSTRUCTION / PROCEDURE

1. Collect all components and verify ratings.
2. Select an isolated low-voltage transformer.
3. Identify diode anodes and cathodes.
4. Connect the four diodes in bridge configuration.
5. Connect the transformer secondary to the bridge AC terminals.
6. Connect the electrolytic capacitor across the DC output.
7. Connect the load resistor across the output.
8. Connect the LED through a series resistor.
9. Check polarity, continuity, and shorts.
10. Apply low-voltage power.
11. Measure AC input and DC output.
12. Observe LED and waveform.
13. Record readings and switch off.



Full-wave bridge rectifier with capacitor-filter output

Safety: Do not connect the breadboard directly to 230 V AC mains.

OBSERVATION AND RESULT

Fill the following table with actual laboratory readings. Do not replace measured values with guesses.

SR. NO.	PARAMETER	OBSERVED VALUE
1	Transformer secondary AC voltage	_____
2	Unfiltered DC output	_____
3	Filtered DC output	_____
4	Load resistance	_____
5	Load current	_____
6	Ripple voltage	_____
7	LED condition	ON / OFF
8	Remarks	_____

Result: The circuit converts low-voltage AC into full-wave pulsating DC. The capacitor reduces ripple and the LED indicates the presence of output voltage.

ADVANTAGES AND LIMITATIONS

ADVANTAGES

1. Both half cycles are used.
2. Higher average DC output than half-wave rectification.
3. Better efficiency than a half-wave circuit.
4. Simple and economical construction.
5. Suitable for low-power electronic circuits.
6. Capacitor filtering produces smoother DC.

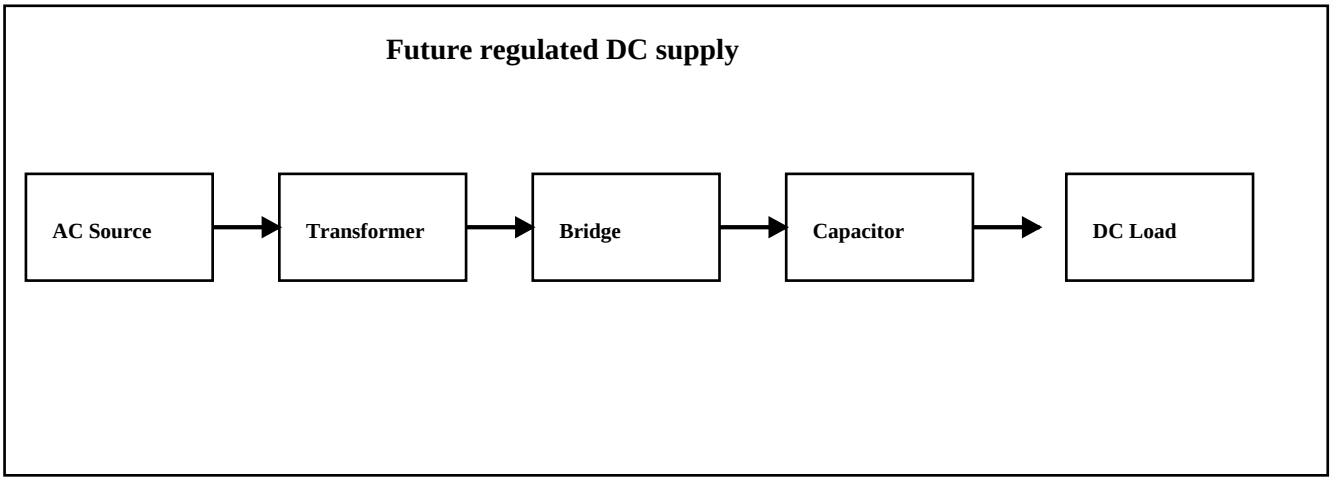
LIMITATIONS

1. Two diodes conduct in series and cause voltage loss.
2. Ripple remains after filtering.
3. Transformer and capacitor ratings are important.
4. Output voltage varies with load.
5. Capacitor polarity must be correct.
6. High-power circuits require thermal and protection design.

CONCLUSION AND FUTURE DEVELOPMENT

The project demonstrates AC to DC conversion using a four-diode full-wave bridge rectifier. The transformer reduces AC voltage, the diodes perform rectification, and the capacitor filter reduces ripple. The resistor provides a load and the LED indicates DC output.

Future development may include a Zener or IC voltage regulator, improved multi-stage filtering, PCB construction, fuse protection, short-circuit protection, and digital voltage-current monitoring.

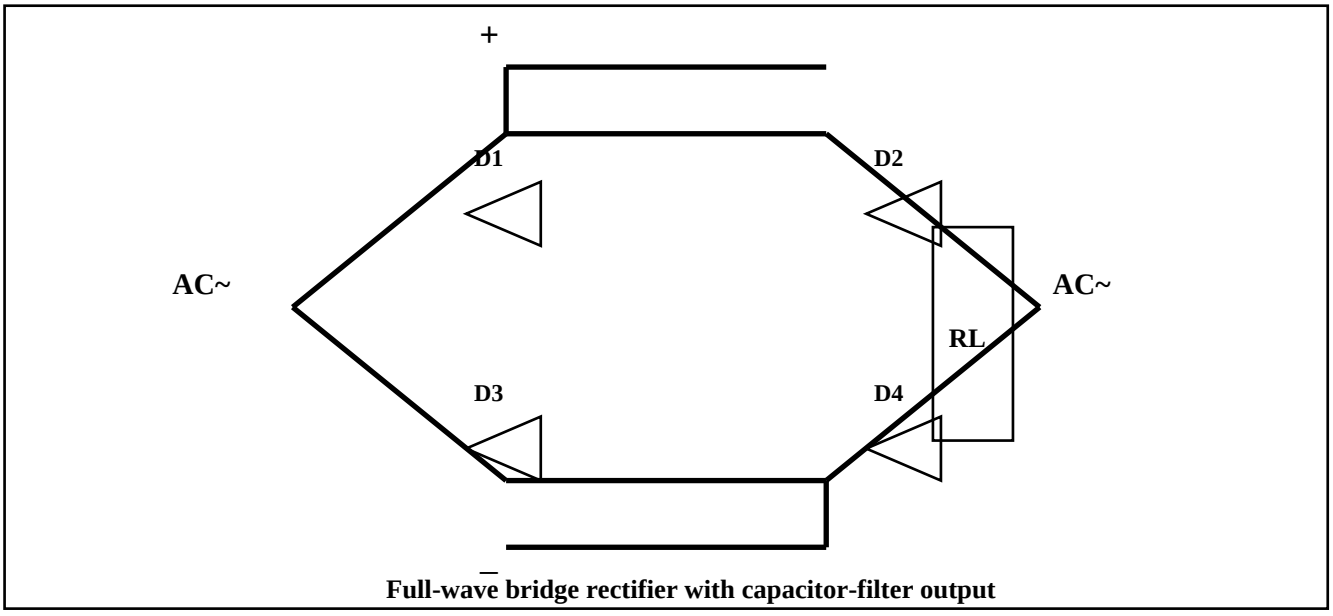


TOTAL COST ESTIMATION

SR. NO.	COMPONENT	COST (₹)
1	Step-down transformer	200
2	PN junction diodes (4)	20
3	Electrolytic capacitor	20
4	Load resistor	10
5	LED and series resistor	10
6	Breadboard / PCB	120
7	Connecting wires	50
8	Switch	15
9	Miscellaneous	75
TOTAL		520

Note: Prices are illustrative and may vary by supplier. Multimeter and laboratory equipment are excluded if already available.

FINAL PROJECT SUMMARY



Full-wave bridge rectifier with capacitor-filter output

A full-wave bridge rectifier with capacitor filter is a simple and useful AC to DC conversion circuit. Four diodes use both half cycles of AC, while the capacitor reduces ripple. The project develops practical knowledge of semiconductor devices, circuit construction, measurement, and electrical safety.

Project Title: **FULL WAVE RECTIFIER WITH CAPACITOR FILTER**
College: [COLLEGE NAME]
Department: [DEPARTMENT NAME]
Students: [STUDENT NAMES]
Guide: [PROJECT GUIDE NAME]

END OF PROJECT REPORT